Part 1:

Question 1.

A DBMS is software that manages data in a structured format, overcoming issues of traditional file systems like data redundancy, poor security, no multi-user support, and no backup.
Example: In a banking system, if two users withdraw from the same account simultaneously, a traditional file system would cause data corruption. A DBMS prevents this using concurrency control and transaction management.

Question 2.

A strong entity has its own primary key. A weak entity lacks a primary key and instead has
a partial key . Its full identifier is formed by combining the partial key with the
primary key of its owner entity.

Question 3.

Phone number should be modeled as a multivalued attribute, since a student can have more than one phone number. A simple attribute holds only one value, and a composite attribute has sub-parts neither applies here.

Question 4.

A super key uniquely identifies a row in a table. A candidate key is a minimal super key no attribute can be removed while still maintaining uniqueness.
Candidate keys: roll_no, email
Super keys: roll_no, email, (roll_no, name), (email, branch), (roll_no, email), etc. Every candidate key is also a super key, but not vice versa.

Question 5.

The Foreign Key constraint is violated. It ensures referential integrity by preventing references to non-existent records. The DBMS will reject and abort the insert operation.

Question 6.

A bridge (junction) table must be created for M:N relationships. Its primary key is a composite of the primary keys of both participating tables. Storing this as a column in one
table would violate 1NF, which requires atomic values.

Part 2.

Part A: Entities and Relationships

Entities:
Customer: customer_id (PK), customer_name, phone_number (multivalued), address (composite: door_no, street/area, pincode)
Order: order_id (PK), delivery_address, price, order_details, quantity, customer_id (FK), restaurant_id (FK)
Restaurant: restaurant_id (PK), restaurant_name
Menu Item: item_id (PK), item_name, price
Relationships:
1. Customer places Order — 1:N
2. Restaurant receives Order — 1:N
3. Restaurant offers Menu Items — 1:N
4. Order contains Menu Items — M:N (resolved via OrderItem bridge table)

Part B: Special Attributes

Multivalued: phone_number — a customer can have multiple phone numbers.
Composite: address — sub-attributes: door_no, street/area, pincode.

Part C: Bridge Table

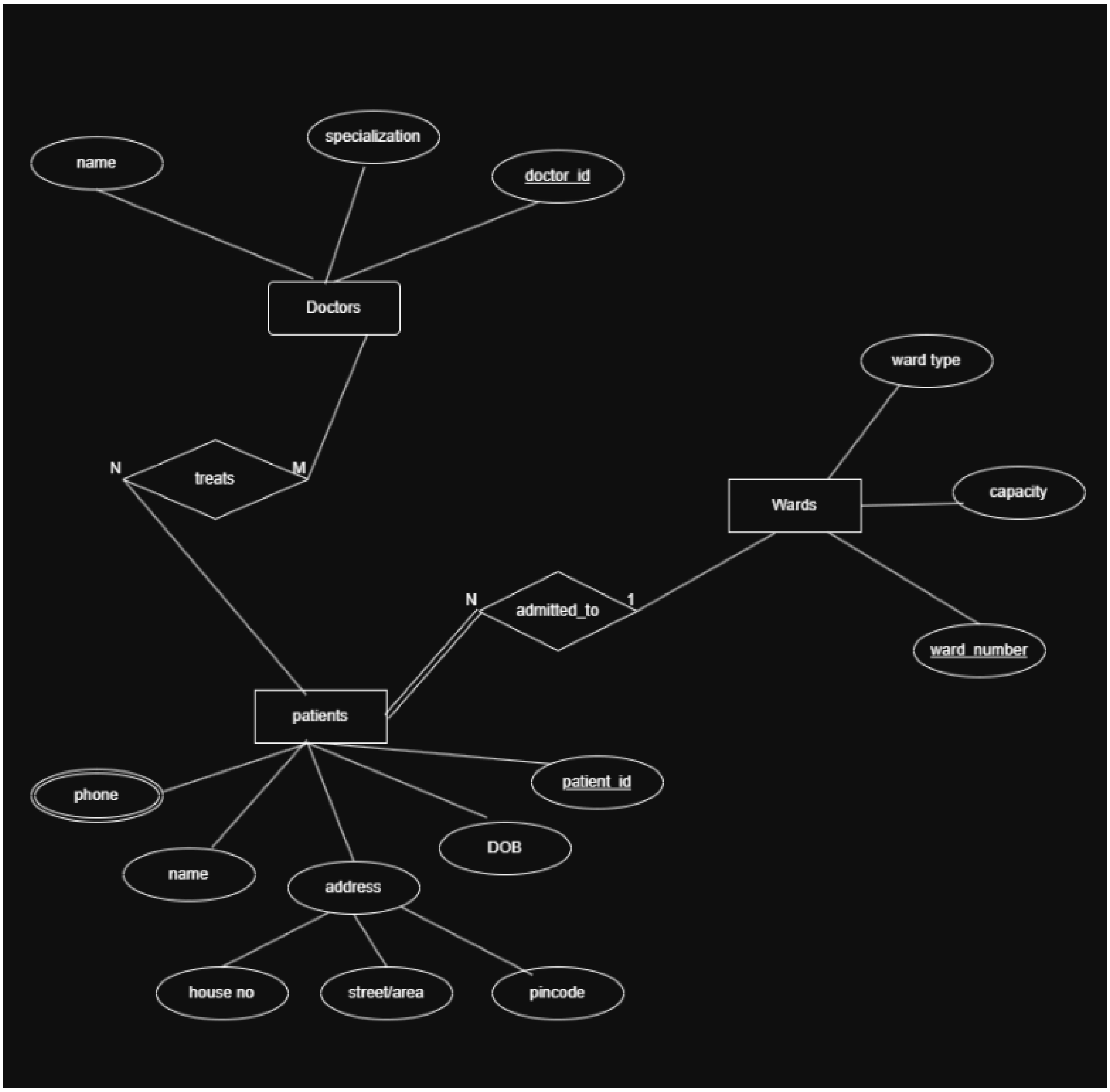
OrderItem(order_id, item_id, quantity)
Primary Key: (order_id, item_id)
Foreign Keys: order_id → Order; item_id → MenuItem
Additional attribute: quantity

Part 3.

Question 2.

The Doctor–Patient relationship is M:N (a doctor treats many patients, a patient sees many
doctors). It is resolved using a Treatment associative entity with composite primary key {doctor_id, patient_id}.
Participation constraints:- Patient has total participation in admitted_to — every patient must be assigned to a ward.
- Ward has partial participation a ward may temporarily have no patients.
- Both Doctor and Patient have partial participation in treats a new doctor may have no patients yet, and a patient record may exist before treatment starts.

address is a composite attribute (house_no, street/area, pincode) phone is multivalued as
patients may provide emergency contact numbers.



name
specialization
doctor_id
Doctors
ward type
N
treats
M
Wards
capacity
N
admitted_to
1
ward_number
patients
phone
patient_id
DOB
name
address
house no
street/area
pincode